

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons) — Semester VI Examination, 2022-23

Paper No. MTB-608: Global Differential Geometry

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 4 marks each.)

- (a) Let $\gamma(s)$ be a unit speed plane curve given by $\gamma(s) = (x(s), y(s))$. Prove that the curvature of the curve $\gamma(s)$ is $|x'y'' - x''y'|$, where prime denotes derivative with respect to arc length s . [3]
- (b) Consider a closed curve in the plane with length equal to 6 cm and bounding an area of 3 cm². Does such a curve exist? Justify your answer. [3]
- (c) Give the definition of a simple curve. Prove that straight lines in $\mathbb{R}^n$ are simple curves.
- (d) Give the definition of a simply connected domain. Also write down the statement of the Jordan curve theorem.
- (e) Give the definition of an open hemisphere with pole N . Prove that the tangent spherical image of a regular closed curve α does not lie in any open hemisphere.
- (f) Give the definition of a compact surface. Let r be the minimum number such that $M \subset B_r(0)$. Prove that $M \cap S_r \neq \emptyset$, where S_r is the sphere of radius r centered at the origin.

Question 2.

- (a) Give the definition of a closed curve. Show that the ellipse $\gamma(t) = (a \cos t, b \sin t, 0)$, $0 \leq t \leq 2\pi$ and $a \neq b$, is a closed simple curve. [6]
- (b) State and prove the Rotation Index Theorem. [6]

Question 3.

- (a) Let A be the area of the region bounded by a simple closed regular plane curve γ of length L . Prove that $L^2 \geq 4\pi A$. [6]
- (b) Prove that the four-vertex theorem is false if the hypothesis “simple” is omitted by considering the curve $\gamma(t) = (\cos t - 2 \sin t \cos t, \sin t - 2 \sin^2 t, 0)$, $0 \leq t \leq 2\pi$. [6]

Question 4.

- (a) If α is an oval of constant width w , then prove that the length of α is πw . [6]
- (b) Prove that the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with semi-axes $a > b > 0$ is an oval. [6]
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Question 5.

- (a) State Fenchel's theorem and verify it for the circle $\alpha(t) = (\cos t, \sin t, 0)$, $0 \leq t \leq 2\pi$. [6]
- (b) If α is a simple knotted regular closed curve in $\mathbb{R}^3$, then prove that the total curvature of α is at least 4π . [6]
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Question 6.

- (a) Define the Weingarten map (or shape operator) L . Prove that the derivative of the unit normal vector is given by $L(X_i) = -D_{X_i}N = \sum h_i^j X_j$, where h_i^j are the components of the shape operator. [6]
- (b) Give the definition of a triangulation of a surface S . State and prove the Gauss-Bonnet theorem for a compact surface. [6]
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Question 7.

- (a) Prove that the total torsion $\oint \tau ds$ of a closed unit speed curve $\alpha(s)$ on the sphere S^2 is zero. [12]